Vol 4 Chapter 9 — The Cosmological Cycle and Interstasis

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Chapter 9 — The Cosmological Cycle and Interstasis

The standard cosmological picture has the universe begin with the Big Bang and expand thereafter, with the cosmological constant Λ driving accelerated expansion at late times. The framework's substrate-mechanism reading goes further: BST predicts a *cosmological cycle hypothesis* with successive Big Bangs separated by periods of substrate equilibration that the team calls Interstasis.

9.1 The D_IV^5 Rigidity Principle

Lyra T2467 + T2468 (May 2026, Casey-named principle #7): two D_{IV}^5 patches in causal information-exchange contact merge into one substrate via Bergman 1922 + Wallach 1976 + Faraut–Koranyi 1994 rigidity. Multi-instance D_{IV}^5 is a category error.

This closes the multiverse loophole structurally. The substrate is *one* — not many instances. What appears as cosmological evolution is the substrate's annealing toward its equilibrium Λ-saturated state.

9.2 The cosmological cycle hypothesis

The substrate's annealing dynamics, integrated across cosmological timescales, produce a cyclic structure: the substrate evolves through a Big Bang phase (rapid commitment-cycle outputs producing high-energy matter), through an expansion phase (commitment cycles diluting), toward an equilibrium Λ-saturated state (Interstasis), and eventually back through another Big Bang.

The period between Big Bangs — Interstasis — is the substrate's equilibration interval. During Interstasis, the substrate's commitment-cycle outputs are minimized; what remains is the substrate's vacuum-Casimir energy density, which is the observed cosmological constant.

9.3 Observational signatures

The cosmological-cycle hypothesis predicts specific observational signatures that distinguish it from standard cosmology:

- Subtle CMB anisotropies at the largest scales reflecting the substrate's previous-cycle imprint
- Cosmic-ray spectrum features at the substrate's natural energy scales
- Gravitational-wave background signatures from previous-cycle Big Bang events

Multi-month research (Task #267) is developing the per-signature substrate-mechanism predictions.

9.4 What comes next

Chapter 10 — dark energy and dark matter — derives the dark sector observables from substrate primaries.

Where to look this up: D_IV^5 Rigidity: T2467 + T2468 (Lyra Friday May 22, 2026). Casey-named principle #7. Cosmological cycle: Task #267 ongoing. For standard cyclic-cosmology models: Steinhardt and Turok, *Endless Universe*.